

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

The number of roots of the equation,

$$(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$$

in the interval $[0, \pi]$ is equal to :

(1) 3

(2) 4

(3) 8

(4) 2

Q2: 17 March (Shift 1) - Single Correct

The inverse of $y = 5^{\log x}$ is :

(1) $x = (1/y)^{\log 5}$

(2) $x = y^{\frac{1}{\log 5}}$

(3) $x = 5^{\log y}$

(4) $x = 5^{\frac{1}{\log y}}$

Q3: 18 March (Shift 1) - Single Correct

The real valued function $f(x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{x - [x]}}$, where $[x]$ denotes the greatest integer less than or equal to x , is

defined for all x belonging to:

(1) all reals except integers

(2) all non-integers except the interval $[-1, 1]$

(3) all integers except 0, -1, 1

(4) all reals except the Interval $[-1, 1]$

Q4: 18 March (Shift 1) - Single Correct

If the functions are defined as $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1 - x}$, then what is the common domain of the following functions:

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$f + g, f - g, f/g, g/f, g - f$ where $(f \pm g)(x) = f(x) \pm g(x), (f/g)(x) = \frac{f(x)}{g(x)}$

(1) $0 \leq x \leq 1$

(2) $0 \leq x < 1$

(3) \$0

(4) \$0

Q5: 18 March (Shift 2) - Single Correct

Let $f : \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$ be defined by $f(x) = \frac{x-2}{x-3}$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be given as $g(x) = 2x - 3$. Then, the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to

(1) 7

(2) 2

(3) 5

(4) 3

Answer Key**Q1 (2)****Q2 (3)****Q3 (2)****Q4 (3)****Q5 (3)**

Hints and Solutions

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Q1 (2)

$$(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$$

$$(81)^{\sin^2 x} + \frac{(81)^1}{(18)^{\sin^2 x}} = 30$$

$$(81)^{\sin^2 x} = t$$

$$t + \frac{81}{t} = 30$$

$$(t - 3)(t - 27) = 0$$

$$(81)^{\sin^2 x} = 3^1 \quad \text{or} \quad (81)^{\sin^2 x} = 3^3$$

$$3^{4 \sin^2 x} = 3^1 \quad \text{or} \quad 3^{4 \sin^2 x} = 3^3$$

$$\sin^2 x = \frac{1}{4} \quad \text{or} \quad \sin^2 x = \frac{3}{4}$$



Q2 (3)

$$y = 5^{\log x}$$

$$y = x^{\log 5}$$

$$y^{\frac{1}{\log x}} = x$$

Repeating $x \rightarrow y$ and $y \rightarrow x$

Q3 (2)

$$f(x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{|x|}}$$

$$\text{Domain} \in (-\infty, -1] \cup [1, \infty)$$

$$\{x\} \neq 0 \text{ so } x \neq \text{integers}$$

Q4 (3)

$$f(x) + g(x) = \sqrt{x} + \sqrt{1-x}, \text{ domain } [0,1]$$

$$f(x) - g(x) = \sqrt{x} - \sqrt{1-x}, \text{ domain } [0,1]$$

$$g(x) - f(x) = \sqrt{1-x} - \sqrt{x}, \text{ domain } [0,1]$$

Hints and Solutions

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$$\frac{f(x)}{g(x)} = \frac{\sqrt{x}}{\sqrt{1-x}}, \text{ domain } [0,1)$$

$$\frac{g(x)}{f(x)} = \frac{\sqrt{1-x}}{\sqrt{x}}, \text{ domain } (0,1]$$

So, common domain is (0,1)

Q5 (3)

$$f(x) = y = \frac{x-2}{x-3}$$

$$\therefore x = \frac{3y-2}{y-1}$$

$$\therefore f^{-1}(x) = \frac{3x-2}{x-1}$$

$$\& g(x) = y = 2x - 3$$

$$\therefore x = \frac{y+3}{2}$$

$$\therefore g^{-1}(x) = \frac{x+3}{2}$$

$$\therefore f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$$

$$\therefore x^2 - 5x + 6 = 0$$

$\therefore$ sum of roots

$$x_1 + x_2 = 5$$